

Graph learning for region materialization in exact shortest-path queries

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Abstract

Road networks answer exact shortest-path queries over and over, yet building an index for the whole network is expensive. This paper studies how to select, under a fixed budget, a small set of mutually disjoint regions for preprocessing, using only historical origin–destination (OD) demand and the static road graph. A parameter-free bidirectional graph diffusion first estimates region value; a graph neural network then refines the ranking, and a budget-aware variant jointly weighs gain, index cost, and region conflicts. The models only select regions offline — online queries are still answered by exact algorithms. Experiments on Porto and Chicago show that the neural model improves global ranking across multiple seeds and two time windows in both cities; the budget-aware version stably reduces query workload in Porto and stays within 3% of the parameter-free method in Chicago. All query distances are 100% correct, and C++ tests show that high-workload queries benefit most. The method improves index resource allocation under a limited budget, while also showing that a better global ranking does not equal faster queries — the two must be verified separately.

Keywords: Shortest paths, Road networks, Workload-aware indexing, Graph learning, Learning to rank, Region materialization

1. Introduction

Shortest-path queries support traffic simulation, travel analysis, and location-based services. Exact graph search can always use the latest road network, but under heavy repeated querying it burns compute continuously; a full index reduces online search at the price of higher preprocessing, storage, and maintenance costs (Dijkstra, 1959; Efentakis et al., 2014). Real demand is far from uniform, so the question that matters in practice is not whether to build an index, but which road regions deserve the limited resources first.

This paper is organized around one main question: under fixed preprocessing and storage budgets, can a system learn — from historical OD endpoints and the static road graph alone — to select a few mutually disjoint regions worth materializing, keeping distances exact while lowering the cost of future queries? The question ties demand prediction, index resource allocation, and exact querying together, and it forces a model to face unseen regions, future workloads, and real deployment constraints rather than merely scoring well against offline labels.

Workload-aware indexing and query caching have shown that historical load can guide shortest-path acceleration (Wan et al., 2022; Dan et al., 2025; Thomsen et al., 2012; Li et al., 2020b). But those methods manage node labels, complete paths, or whole-graph partitions, and they do not directly answer how a handful of regions should be ranked between cost and future gain. We adopt a mature exact region-overlay structure as the system substrate and narrow the research to demand-driven region value estimation, in-budget set selection, and leakage-free end-to-end validation.

To answer the main question, we first build a label-free region ranking from bidirectional multi-scale demand propagation, then let a deep graph network learn the ranking residual beyond this frozen prior, and finally optimize the gain at the budget truncation point jointly with index cost and candidate conflicts. Artificial intelligence is confined to offline resource allocation: it does not predict shortest-path distances and never enters the user request path; online queries continue to be answered by exact graph algorithms. This design keeps the model’s increment independently auditable, and lets a system refresh its materialized regions periodically without replacing its existing query service.

The two-city experiments return a three-layer answer. First, the neural residual improves the full candidate ranking consistently in 12 paired comparisons spanning two cities, three seeds, and two external windows, evidence that it captures repeatable information beyond deterministic propagation. Second, budget-aware training stably protects the head of the ranking while cutting index cost: Porto gains a cross-seed consistent online workload improvement, and Chicago stays within 3% of the strong prior. Third, all query distances are 100% correct, and same-implementation C++ benchmarks together with query-scale analysis show that high-workload queries best amortize region access costs — and that global ranking, the budget head, and real deployment gain must each be verified on their own.

Our contributions are:

1. We define the demand-driven selective region materialization task, measuring region value by the reduction in exact query workload, and give a shadow label-acquisition and periodic deployment pipeline that stays

off the online critical path.

2. We propose a layered method — parameter-free demand propagation, a neural ranking residual, and budget-aware joint optimization — so that the deterministic prior, the learned increment, and the deployment constraints can each be verified separately.
3. We establish a verification loop of temporal isolation, spatial overlap-group isolation, strictly disjoint selection, and exact online pairing, providing multi-seed, two-city evidence that runs from offline ranking to real system performance.
4. We empirically separate the full candidate ranking, the budget head, index cost, and set-level query gain, and show the engineering value of a high-quality global ranking for dynamic budgets, conflict-driven elimination, and network-wide stand-by allocation.

The experiments answer the main question along a single evidence chain: first verify that region value is predictable and generalizes to the future, then confirm the effective signal by ablation, then test disjoint deployment, wall-clock performance, and budget-aware optimization, and finally analyze the applicability boundary set by spatial position and query scale. The seven links share one candidate pool, one budget, and one exactness constraint, and each supplies a piece of evidence the main conclusion cannot do without.

2. Related work

2.1. Exact shortest-path indexes and region overlays

Exact shortest-path acceleration reduces online work mainly through search pruning, labeling indexes, and partition-based overlays. A partition overlay summarizes paths inside each region into boundary connections and preserves exact distances on the compressed graph; SALT unifies this structure with several classes of shortest-path queries (Efentakis et al., 2014). This line of work concentrates on how an index is built and queried, and usually assumes the partitioning scheme and the materialization scope are given. We adopt the mature structure directly as our substrate and study what happens when a limited budget permits only a few regions: how should future usage value decide where the resources go? The underlying overlay guarantees that answers are exact; our ranking and selection decide what gets materialized.

2.2. Workload-aware indexing and shortest-path caching

Workload-aware indexing exploits the spatial skew and temporal locality of query distributions to give frequently used objects more index resources. WCF adjusts an exact labeling index by historical query workload (Wan et al., 2022), and FAHL folds road-flow information into label construction (Dan et al., 2025). Both show that demand information improves exact querying, but the objects they manage are node labels or index cores. Our problem is materialization choice inside a region overlay: every candidate simultaneously carries a gain, a boundary connection

cost, and conflicts with other candidates, so node frequency cannot stand in for the decision.

Shortest-path caching uses history from another angle. Path caches keep complete paths likely to recur, and regionalized caches also weigh frequency, computation cost, and spatial coverage (Thomsen et al., 2012; Li et al., 2020b). Caching is suited to reusing objects already computed, whereas we want to configure exact region indexes for queries that have not yet arrived — without reading historical shortest paths. The two share the idea of workload-driven allocation of limited resources, but differ in input information, materialized objects, and how work is reused online. We therefore exclude historical paths from the model input outright and hold the final test to the exact workload of future queries.

2.3. Traffic graph learning and multi-scale propagation

Traffic graph learning commonly uses road direction and neighborhood propagation to predict flow or speed. DCRNN expresses bidirectional spatial dependence on roads with directed random walks (Li et al., 2018), and SIGN together with Jumping Knowledge shows that multi-scale representations help combine different propagation ranges (Frasca et al., 2020; Xu et al., 2018). These works predict traffic state; they do not allocate region budgets for exact query indexes. We borrow the mature propagation ideas to map historical origin and destination demand to region value, and use the layered design of a parameter-free prior plus a supervised residual to separate fixed propagation from learnable information.

2.4. Deep graph networks and ranking optimization

Deep graph networks need stable signal and gradient paths. NBFNet organizes graph reasoning around message propagation (Zhu et al., 2021), while residual connections and cross-layer readouts are general tools against deep-training difficulty (Li et al., 2020a; Touvron et al., 2021). Learning-to-rank research further supplies differentiable approximations of order, letting a training objective attend directly to the relative ranking of candidates (Jagerman et al., 2022). These components give us the methodological basis, but region materialization additionally demands handling the full ranking, budget truncation, index cost, and spatial conflicts at once. Our focus is organizing these objectives into a deployable pipeline for exact road queries, and testing under frozen protocols whether the neural increment truly converts into engineering gain.

3. Problem definition

3.1. Road graph, query workload, and time windows

Let $G = (V, E, w)$ be a directed road graph with positive edge weights $w : E \rightarrow \mathbb{R}_{>0}$ giving road lengths. A query $q = (o_q, d_q, t_q)$ carries an origin, a destination, and a timestamp. All queries, sorted by time, are cut into three consecutive, non-overlapping windows: a history window $H =$

$[0, 0.35)$, a current label window $Y = [0.35, 0.70)$, and a frozen future window $F = [0.70, 1.00)$. H only builds model inputs; Y supplies training labels and train/validation selection; F is opened once, after the protocol freeze, for zero-shot temporal evaluation.

From H we keep only endpoint marginals. For a node v ,

$$\begin{aligned} h_O(v) &= \frac{1}{|H|} \sum_{q \in H} \mathbf{1}[o_q = v], \\ h_D(v) &= \frac{1}{|H|} \sum_{q \in H} \mathbf{1}[d_q = v]. \end{aligned} \quad (1)$$

No method reads the shortest path of any query in H .

3.2. Candidate regions and exact gain labels

The candidate set $\mathcal{R} = \{R_1, \dots, R_M\}$ is generated over the whole graph with a fixed random seed, as spatially uniform as possible; each candidate is a connected subgraph of 512 nodes. Candidate generation reads no historical OD data, no labels, and no model output. The boundary B_R of a region R contains its nodes adjacent to the outside; the rest form the interior I_R .

For a label query q , let $N_G(q)$ be the number of nodes expanded by bidirectional Dijkstra on the original graph, and $N_R(q)$ the total expansion of the overlay search plus local endpoint access when region R alone is materialized. The supervised label of R is

$$y_R^Y = \frac{1}{|Q_Y|} \sum_{q \in Q_Y} [N_G(q) - N_R(q)], \quad (2)$$

where Q_Y holds 2,000 queries drawn from Y with a pre-registered seed. The future label y_R^F uses the same definition but is generated from F only at final evaluation. The label measures exact-algorithm workload rather than noisy single-run Python wall-clock time, and every query additionally checks that the indexed distance equals the original distance.

Label acquisition in production, and why deployment stays light.. Equation (2) is a counterfactual workload under a single-region overlay. It usually cannot be recovered from existing shortest-path logs, but it also requires no synthetic OD load. A production system can mirror a small share of naturally arriving queries to asynchronous shadow workers, which load candidate overlays in rotating batches or parallel shards and record, per paired query, $N_G(q)$, $N_R(q)$, scanned edges, latency, distance agreement, and index version. Once samples cross a threshold they aggregate into region labels. The live service keeps answering from the current stable index; neither shadow execution nor the neural model touches the user request path, and after training the production side only receives a candidate ranking and a materialization plan on a fixed cadence. Engineering cost therefore reduces to bypass sampling, offline statistics, and periodic index maintenance — no online shortest-path code

is replaced by neural inference. The protocol adds no OD-generation cost and can be rate-limited, scheduled off-peak, paused, or parallelized, though candidate query computation, overlay storage, and audit logs are not strictly free. Our study — two cities, two windows, 1,200 candidates per city, 2,000 queries per candidate, 9.6 million candidate-query evaluations in total — can be read as a controlled, repeatable offline simulation of this shadow protocol on a fixed query set.

3.3. Selective materialization objective

A ranker outputs a score s_R per candidate. Deployment greedily picks $K \in \{5, 10, 18\}$ regions in descending score order, subject to pairwise node-disjointness:

$$|S| = K, \quad R_i \cap R_j = \emptyset, \quad \forall R_i \neq R_j \in S. \quad (3)$$

The hard-disjoint constraint comes from the materialization semantics of the index itself. Because Equation (2) is a marginal single-region gain, gains of a set do not add; the value of S is settled only by materializing the whole set at once and running real online queries against it.

4. Method

4.1. CRP-style selective region materialization

For each selected region R , the offline stage runs restricted Dijkstra on the induced subgraph $G[R]$ from every $u \in B_R$ and adds, for each reachable $v \in B_R$, a directed shortcut

$$w_R(u, v) = d_{G[R]}(u, v). \quad (4)$$

It then removes I_R , keeping outside nodes, boundary nodes, surviving original edges, and the shortcuts. The construction follows the established CRP/boundary-overlay idea (Efentakis et al., 2014). When a query endpoint falls inside some I_R , the algorithm computes exact distances from that endpoint to the region boundary only, seeds the bidirectional search frontier with them, and, when both endpoints share a region, also compares the direct in-region distance. A query never falls back to the original graph, and exactness is preserved.

Distance exactness.. Any original-graph path crossing region R can be cut at the boundary nodes where it enters and leaves R . Replacing each in-region subpath by the corresponding shortcut cannot lengthen the path, since the shortcut stores the shortest distance within $G[R]$; conversely, every shortcut corresponds to a real path in $G[R]$, so the overlay cannot produce an illegally short distance. Interior endpoints enter through exact in-region frontiers, and same-region pairs additionally compare the direct in-region path. With selected regions pairwise disjoint, shortcuts exact, and edge weights non-negative, overlay distances equal original-graph distances. The implementation still verifies this per query; the argument does not stand in for the engineering test.

Preprocessing and query cost. Write b_R for the boundary size of R and m_R for its induced interior edge count. Running restricted Dijkstra from every boundary node costs about $O(b_R(m_R + |R|) \log |R|)$, with $O(b_R^2)$ potential shortcuts. An identical 512-node budget therefore does not imply identical index cost: regions with denser boundaries emit more shortcuts and more online edge scans. We report region node budget, shortcut count, expanded nodes, and scanned edges side by side precisely so that equal node counts are not mistaken for equal system cost.

4.2. Z0: a parameter-free bidirectional multi-scale demand field

Historical marginals are first log-scaled:

$$\mathbf{o}^{(0)} = \log(1 + 1000 \mathbf{h}_O), \quad \mathbf{d}^{(0)} = \log(1 + 1000 \mathbf{h}_D). \quad (5)$$

Let $P_{\rightarrow}$ be the receiver-normalized mean-propagation matrix along original edge directions and $P_{\leftarrow}$ its counterpart on the reverse graph. Each layer applies a fixed lazy mean with a self-loop:

$$\begin{aligned} \mathbf{o}^{(k+1)} &= \frac{1}{2}(I + P_{\rightarrow}) \mathbf{o}^{(k)}, \\ \mathbf{d}^{(k+1)} &= \frac{1}{2}(I + P_{\leftarrow}) \mathbf{d}^{(k)}. \end{aligned} \quad (6)$$

Over the depth set $\mathcal{D} = \{1, 2, 4, 8, 16, 32\}$, the region score is

$$\begin{aligned} z_R &= \frac{1}{|\mathcal{D}|} \sum_{k \in \mathcal{D}} \left[\text{mean}_{v \in R} e_v^{(k)} + \max_{v \in R} e_v^{(k)} \right], \\ \mathbf{e}^{(k)} &= \mathbf{o}^{(k)} + \mathbf{d}^{(k)}. \end{aligned} \quad (7)$$

Z0 has no learnable parameters and reads neither Y nor F . It does not try to predict paths; it estimates how strongly a candidate region is exposed to historical origin and destination demand across multi-scale road neighborhoods.

4.3. BRIDGE: a neural ranking residual on the Z0 prior

BRIDGE runs an origin tower on the forward graph and a destination tower on the reverse graph, with no parameter sharing. Each layer applies an edge-feature-gated message-aggregate branch $\mathcal{F}_\ell$ inside a pre-norm update whose skip path is a strict identity:

$$\mathbf{h}^{(\ell+1)} = \mathbf{h}^{(\ell)} + 0.01 \mathcal{F}_\ell(\text{LayerNorm}(\mathbf{h}^{(\ell)}), E). \quad (8)$$

This update exists to suppress the exponential backward amplification observed in the earlier 32-layer backbone; it is a stabilization of known deep-residual ideas for this task, not a new residual architecture. The model reads both towers at depths 1, 2, 4, 8, 16, 32, applies region mean+max pooling, and outputs a residual $r_\theta(R)$. The final score is

$$s_R = z_R + \alpha r_\theta(R), \quad \alpha \in \{0, 0.125, 0.25, 0.5, 1\}, \quad (9)$$

where $\alpha = 0$ falls back to Z0 exactly. Each epoch’s α is chosen on validation only, with ties resolved toward the smaller residual.

4.4. A soft Spearman objective aligned with evaluation

The earlier pairwise softplus loss could keep decreasing by inflating score scale while true Spearman moved the other way. To remove this misalignment, batch scores are standardized to $\hat{s}_i$ and soft ranks are built from pairwise sigmoids (Jagerman et al., 2022):

$$\tilde{r}_i = \sum_j \sigma\left(\frac{\hat{s}_i - \hat{s}_j}{\tau}\right), \quad \tau = 0.1. \quad (10)$$

With r_i^* the mean ranks of the true labels, the training loss is

$$\mathcal{L}_{\text{rank}} = 1 - \text{Pearson}(\tilde{\mathbf{r}}, \mathbf{r}^*). \quad (11)$$

Checkpoints are selected by validation Spearman for every model. Holdout and F take no part in choosing structure, learning rate, epochs, α , or strategy.

4.5. BRIDGE-B: budget-aware training

Soft Spearman constrains the full candidate order but says nothing about truncation at a fixed budget, index cost, or region conflicts. BRIDGE-B keeps the same stable 32-layer towers and frozen Z0 prior, fixes the residual gate at 1, and builds differentiable top- K weights inside each training split. For standardized scores $\hat{s}_i$, with t_K the stop-gradient K -th largest score,

$$p_i^{\text{raw}} = \sigma\left(\frac{\hat{s}_i - t_K}{T}\right), \quad p_i = \frac{K p_i^{\text{raw}}}{\text{sg}(\sum_j p_j^{\text{raw}})}, \quad T = 0.15, \quad (12)$$

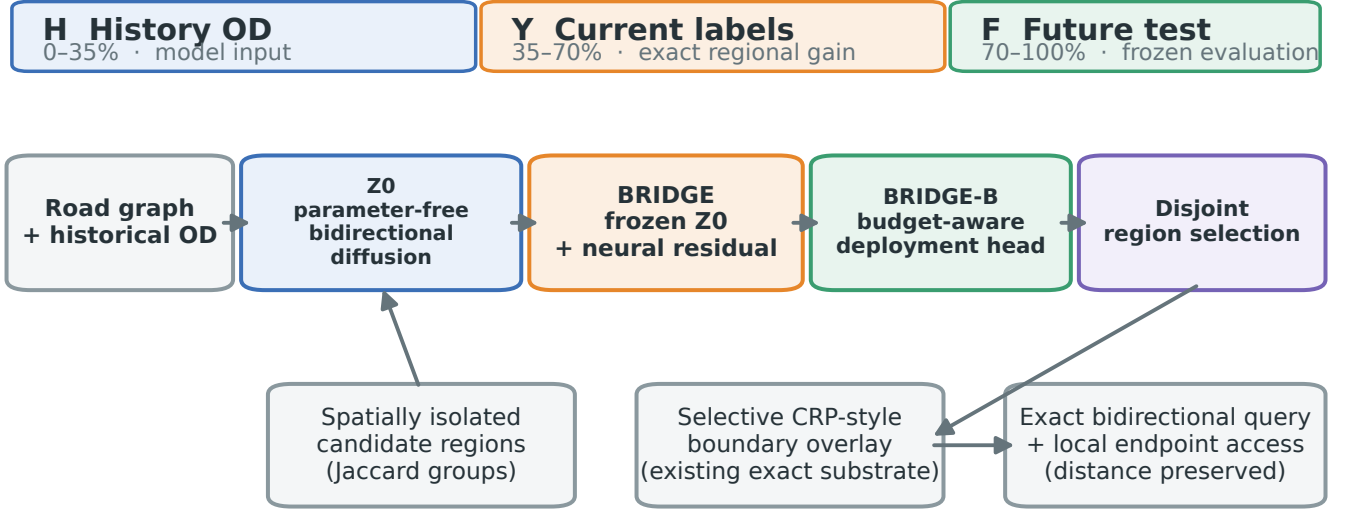
so that the soft selection mass sums to approximately K . Let $\bar{y}_{\text{top}} = \sum_i p_i y_i / \sum_i p_i$, let C_i be the shortcut count of candidate i , and let M_{ij} mark conflicting candidate pairs. The training objective is

$$\mathcal{L}_B = \mathcal{L}_{\text{rank}} - \lambda_g \bar{y}_{\text{top}} + \lambda_b \phi\left(\frac{\sum_i p_i C_i}{B} - 1\right) + \lambda_c \frac{\sum_{i,j} p_i M_{ij} p_j}{(\sum_i p_i)(\sum_i p_i - 1)}, \quad (13)$$

with sg the stop-gradient operator and $\phi(x) = 0.05 \text{softplus}(x/0.05)$. The training budget B is the Z0 soft top- K shortcut cost on the same split; snapshot screening separately uses its hard-disjoint cost. Weights are $\lambda_g = 1.0$, $\lambda_b = 1.0$, $\lambda_c = 0.10$. Full-candidate deployment is 18/1200, and the same ratio gives 13/840 on train and 3/180 on validation.

A validation snapshot is saved every epoch. The final checkpoint is chosen only among updated snapshots whose hard-disjoint validation top- K shortcuts stay within the Z0 budget and whose Spearman stays within 0.005 of Z0, maximizing true top- K mean label gain among the feasible ones. Holdout never participates; full-candidate Y/F online evaluation runs only after the checkpoint freeze. The protocol connects “training a ranking score” with “whether the set can be deployed,” while keeping exact online queries as the final, non-negotiable gate.

Leakage-free temporal protocol



Learning allocates offline materialization resources; online shortest-path answers remain exact.

Figure 1: End-to-end pipeline. The overlay substrate is an existing exact index structure; this paper studies historical-OD-driven region ranking, isolation, and selective materialization. Learning allocates offline resources only — online queries stay exact.

Table 1: Data and frozen splits for the two cities.

City	Trip data	Nodes	Directed edges	Total OD	H	Y	F
Porto	taxi	133,839	221,589	98,082	34,328	34,329	29,425
Chicago	bike share	108,114	200,155	100,000	35,000	35,000	30,000

5. Experimental design

5.1. Datasets and graph scale

Porto uses the public taxi trajectory dataset (Moreira-Matias et al., 2013), mapping the first and last points of valid trajectories to OD pairs. Chicago uses Divvy bike-share trips from June 2022 (Divvy, 2022), cleaned by field validity, temporal order, study boundary, and snap quality, then subsampled to 100,000 trips at equal intervals over the full time series — never filtered by model output or region gain. Road networks come from OpenStreetMap (OpenStreetMap contributors, 2026); Chicago is pinned to the January 2022 Illinois historical snapshot closest to the trip period rather than whichever recent extract scores better. Each city gets 1,200 candidates of 512 nodes, split 840/180/180 into train/validation/holdout. Pairwise Jaccard overlaps at or above 0.50 define edges of an overlap graph, and each connected component is assigned to a single split. Porto and Chicago form 521 and 452 overlap groups respectively; no candidate pair at or above the threshold crosses a split.

Both Y and F label all 1,200 candidates with 2,000 fixed queries each, so the two cities together comprise 9.6 million candidate-query label evaluations. Candidates, query sampling seeds, and splits were frozen before any holdout or future result was read.

5.2. Leakage control and freezing order

Candidate regions are spatial objects, and neighboring BFS regions often share most of their nodes. Under an early random candidate split, predicting a test candidate’s label from its nearest overlapping training candidate reached a Spearman of about 0.9936 — a random split badly overstates generalization. We therefore build the Jaccard overlap graph at threshold 0.50 first and assign whole connected components as the minimal unit. This is stricter than removing exact duplicates, and it blocks the model from “memorizing” a test region through shared nodes.

Temporally, H , Y , and F share no query IDs and no time range. Candidate identity is fixed from the static road graph before H is read; model inputs come from H only; training and validation labels come from Y ; holdout serves spatial external evaluation only; and F is unlocked after structure, hyperparameters, seed set, and selection strategy are all frozen. Chicago reuses the protocol fixed on Porto without re-tuning depth, learning rate, temperature, or residual gates against its own holdout, future, or online results.

This freezing order covers the Z0/BRIDGE main line. BRIDGE-B is a transparent extension carried out after Y/F were already unlocked: stages S0–S2 developed the budget objective on seed 42, and S3 froze the S2 protocol

before adding seeds 43/44, without adjusting seed 44 in response to seed 43. S3 therefore supports repeated-seed stability on the same cities and windows; it does not repack-age the unlocked Y/F as a fresh temporal confirmation, which would require a new future window.

5.3. Baselines

We compare: random ranking; a hotspot method ranking by historical endpoint frequency; a midpoint proxy using only region geometric centers; an MLP without graph propagation; parameter-free Z0; the neural residual BRIDGE; and the budget-aware BRIDGE-B. The MLP reports 5 random seeds; BRIDGE and BRIDGE-B report seeds 42/43/44. All methods share the same candidate pool and deployment budget, and no seed is dropped in response to holdout or future results.

Each baseline isolates one question. Random and hotspot ask whether arbitrary regions or endpoint frequency alone already pay off. The midpoint proxy asks whether plain spatial position suffices. The MLP asks how far supervised region features go without propagating over road adjacency. Z0 tests fixed graph diffusion, and BRIDGE tests only whether a supervised residual beats the strong prior. An oracle appears solely as the label-space ceiling — never as a deployable method, never in hyperparameter selection. BRIDGE-B asks one further question: on the global ranking axis BRIDGE already establishes, can explicit budget objectives improve the fixed- $K=18$ head and its real deployment?

5.4. Metrics and implementation

Offline ranking reports Spearman, $\text{NDCG}@K$, and top- K mean true gain for $K \in \{5, 10, 18\}$. Deployment reports shortcut count, mean expanded-node change, mean and P95 latency, and distance correctness. Online change rates are relative to same-implementation bidirectional Dijkstra on the original graph. BRIDGE has 418,183 parameters and trains with CUDA FP32, learning rate 5×10^{-3} , 40 epochs, no scheduler. BRIDGE-B shares the structure and learning rate at a fixed 12 epochs, again without a scheduler — a deliberately short leash so that training does not scale up before the real deployment gate is passed. All training runs on GPU.

The C++ benchmark implements original-graph and overlay bidirectional Dijkstra inside one C++20/O3 executable, pinned to a single CPU core, with 2 warm-up rounds and 10 timed rounds. Keeping one language and one machine prevents implementation noise from being booked as algorithmic gain.

Neural training uses CUDA on an RTX 4090 D; CPUs handle data audits, unit tests, and the final C++ query benchmark. Means and standard deviations cover the pre-registered seed sets in full. We do not dress comparisons across different seed counts as significance tests, and we do not drop outlier seeds after seeing holdout. Online timing pairs each query’s original-graph and overlay runs

in the same process and order; system-level interference is handled by a full clean rerun, not by selective sample deletion.

AI-assisted software development and verification.. During the research, OpenAI Codex assisted with writing, refactoring, and debugging parts of the Python/C++ code, tests, and experiment scripts; the server-side model is managed by the platform and carries no permanently fixed public version identifier. All changes were reviewed by the authors under local version control and validated through unit tests, fixed-configuration reproduction, per-query original-versus-indexed distance consistency checks, and machine-readable result audits. The tool does not generate or modify experimental observations, does not substitute for formal model training, and takes no part in method selection, result interpretation, or final scientific judgment.

6. Results

This section answers only the single main question posed in the introduction: can historical demand guide exact region materialization under a limited budget? To keep any one metric from standing in for the end-to-end conclusion, the evidence unfolds in order — predictability, temporal generalization, signal source, disjoint deployment, real performance, budget optimization, and applicability boundary.

6.1. Evidence 1: historical demand predicts region value

Z0 beats the midpoint proxy and the MLP on global Spearman in both cities: propagating historical demand over road topology yields a strong region-value axis on its own. On top of it, BRIDGE’s per-seed Spearman increments are $+0.012373/ +0.010313/ +0.008212$ in Porto and $+0.006405/ +0.006515/ +0.004035$ in Chicago — six comparisons, six gains. Consistency of this kind across cities and seeds is hard to explain as noise; the residual is picking up reproducible ordering information that Z0 lacks. Chicago’s $\text{NDCG}@18$ moving in the opposite direction is the counterpart finding: soft Spearman genuinely optimizes the full candidate order, while head quality under a fixed budget answers to a different objective. The two metric families separating is not a failure of the global gain — it is the experiment locating the boundary between two learning targets.

6.2. Evidence 2: frozen rankings generalize to the future window

All three BRIDGE seeds beat their city’s Z0 on future Spearman. Combined with the spatial holdout, BRIDGE wins all 12 paired comparisons across two cities, three seeds, and two external windows — the core stability evidence for the neural gain. One caveat scopes the claim: Porto’s Y/F region-gain Spearman is 0.9959, so the Porto result certifies zero-shot temporal generalization under a stable

Table 2: Spatial holdout ranking. MLP averages five seeds; BRIDGE averages three.

City	Method	Spearman	NDCG@18
Porto	midpoint proxy	0.891756	0.983040
	MLP	0.869552 ± 0.019732	0.959260 ± 0.014273
	Z0	0.935603	0.974533
	BRIDGE	0.945902 ± 0.001699	0.974551 ± 0.000110
Chicago	midpoint proxy	0.895597	0.867650
	MLP	0.847990 ± 0.023984	0.901271 ± 0.028814
	Z0	0.941135	0.922494
	BRIDGE	0.946787 ± 0.001144	0.866699 ± 0.033993

Table 3: Frozen future-window ranking.

City	Method	Spearman	NDCG@5	NDCG@18
Porto	Z0	0.944656	0.944421	0.933698
	BRIDGE	0.956598 ± 0.001580	0.893761 ± 0.005628	0.919343 ± 0.001397
Chicago	Z0	0.897509	0.659842	0.786935
	BRIDGE	0.925712 ± 0.000866	0.697910 ± 0.046648	0.765981 ± 0.030726

demand regime, not robustness to strong distribution drift. Chicago drifts more, and there BRIDGE’s global increment grows — while its top- K still fluctuates.

6.3. Evidence 3: road topology and bidirectional demand form the effective signal

Degree-preserving rewiring keeps every node’s in- and out-degree while destroying real road adjacency. The Spearman collapse across all four city-window cells supports an independent contribution of correct road topology to the global value axis. Origin-only and destination-only variants both fall below bidirectional Z0, with the destination field contributing more. The undirected variant nearly matches Z0 globally but loses future NDCG@5, so direction matters mostly at the head. Re-coupling OD pairs while preserving both marginals barely changes anything — the evidence supports marginal-demand propagation, not the original OD pairing, as the necessary ingredient. Max pooling almost reproduces the full score; mean pooling is clearly weaker. Multi-depth readout is the most robust choice across splits and windows without being the top scorer on every single metric.

6.4. Evidence 4: rankings convert into exact disjoint deployment

Selecting Porto’s top 18 directly by Z0 score yields a union of only 2,773 unique nodes at a member redundancy of 3.323, including one candidate pair sharing 116 nodes — a set the index cannot co-materialize. Hard-disjoint greedy selection fills all 18 mutually disjoint regions covering 9,216 unique nodes, giving up some single-region gain for a set with well-defined deployment semantics. Predicting valuable regions and composing a valuable region *set* are different problems, and this is where they part.

In Porto, BRIDGE shaves only about 0.13/0.04 percentage points of current/future expansions beyond Z0 while carrying about 15.5% more shortcuts, so Z0 keeps the better performance-per-byte. In Chicago, Z0 leads BRIDGE and the proxy outright on expanded nodes. Including BRIDGE-B, the Python online evaluations total 228,000 query-index pairs, every one returning the exact distance. Global ranking gains, in short, become system claims only after passing deployment — not before.

6.5. Evidence 5: query structure decides wall-clock gains

The C++ benchmark covers 800,000 formally timed query pairs at 100% distance correctness, replaying the Python regions and shortcuts exactly. Porto cuts expansions and edge scans together, so both mean and P95 latency fall. Chicago cuts expansions by roughly 20% yet scans 18–28% more edges through its dense boundary shortcuts, which cancels the gain for typical queries; even so, 14 of its 20 method-window combinations improve P95. Whether materialization speeds a city up thus depends on boundary shape and shortcut density, not on ranking accuracy alone.

6.6. Evidence 6: a budget-aware objective stably coordinates head and cost

Before training BRIDGE-B, we test whether cost awareness needs training at all. E0 is a zero-training linear post-processor: convert predicted score and per-region shortcut count to percentile ranks, then take $r_i = \text{rank}(s_i) - \lambda \text{rank}(c_i)$ with $\lambda \in \{0.1, 0.3\}$. All 12 fixed configurations (two cities, three seeds) went through the $K=18$, Y/F exact online gate, with no widening of the weight grid afterwards. In Porto, E0 slashes shortcuts from Z0’s 6,712 down

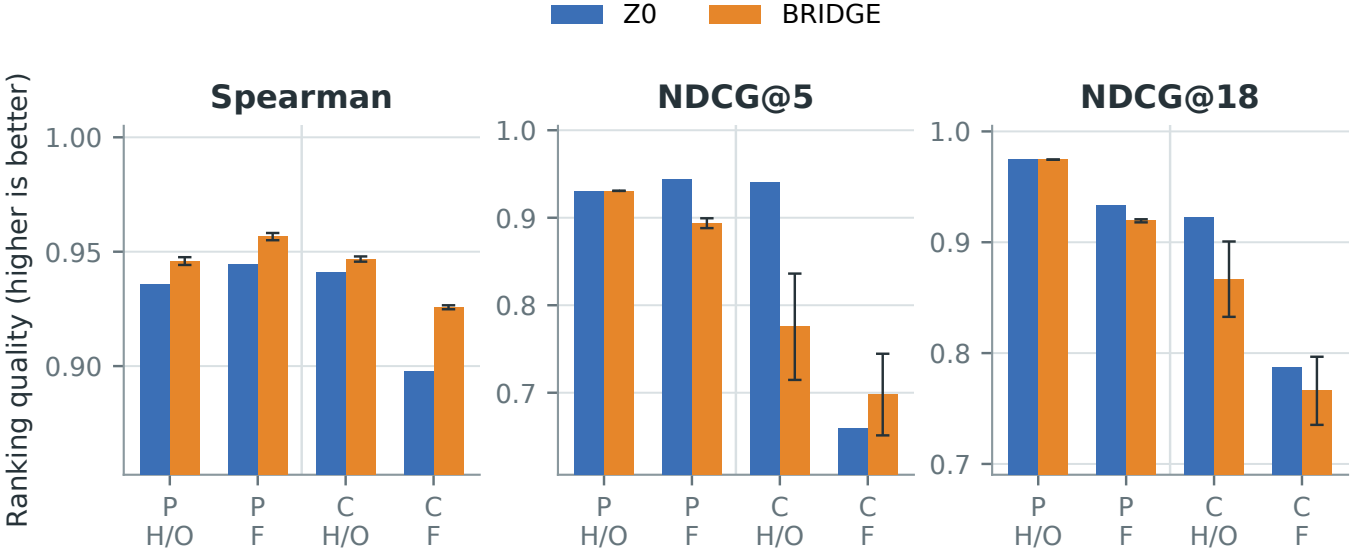


Figure 2: Ranking on the spatial holdout (H/O) and frozen future window (F) in both cities. BRIDGE bars show three-seed means with population standard deviation. Global Spearman rises consistently while head NDCG moves independently — the two training objectives, made visible.

Table 4: Correct road topology versus degree-preserving rewiring (Spearman).

City and window	Z0	Degree-preserving rewiring	Absolute drop
Porto holdout	0.935603	0.556581	0.379022
Porto future	0.944656	0.572467	0.372189
Chicago holdout	0.941169	0.605370	0.335799
Chicago future	0.897844	0.454814	0.443030

to 3,135–4,499, but all six Y/F runs expand roughly 28–110 more nodes per query than Z0 — a plain cost–benefit imbalance. In Chicago, shortcuts drop from 23,378 to 13,049–15,122, yet only seed 43 at $\lambda = 0.3$ edges out Z0 in both windows. One of twelve configurations passes the joint two-window gate. All 24 method–window online results remain 100% distance-correct, so the failure is not in the index — it is that a fixed subtraction after the fact cannot coordinate cost against real deployment gain across cities, seeds, and weights. E0 can cut storage aggressively; as a multi-objective deployment method it is unreliable, and it never reproduces the cross-seed consistency BRIDGE-B reaches in Porto. That contrast is the direct case for joint training.

S0–S2 developed the method on seed 42 under short training. S2 changed exactly one thing — the top- K gain weight in Equation (13), from 0.25 to 1.0 — and first achieved, in Porto, lower shortcuts and lower Y/F expansions at once. S3 then froze the S2 protocol, added seeds 43/44, and ran the full-candidate inference and exact online gate uniformly over all three seeds.

Porto’s three seeds select the same 18 members — only their internal order swaps slightly — so the Y/F expansion figures have zero variance across seeds, and shortcuts drop from Z0’s 6,712 to a stable 6,644. BRIDGE-B holds the

head in Porto and converts the budget objective into a real, repeatable online workload improvement.

Chicago’s seeds 42/43 pick identical members; seed 44 differs in part. All three seeds beat Z0’s 0.941135 hold-out Spearman, all exceed 0.940545 on NDCG@5, and all stay below Z0’s 23,378 shortcuts — but mean Y/F expansions run 65.947/72.032 nodes above Z0. Against Z0’s 2,609.917/2,530.114 expansions per query, that gap is about 2.53%/2.85%; Porto’s improvement is about 0.84%/0.87%. Both cities’ true online workloads are therefore best read as parity with Z0 — Porto slightly ahead, Chicago slightly behind — while global ranking, head quality, and index cost are all under stable control, with expansion differences confined to under 3%. The 24,000 BRIDGE-B online pairs across three seeds are all distance-correct. What the experiment adds is a finer split: “cheaper index,” “higher head gain,” and “fewer set-level expansions” are three related objectives that do not substitute for one another.

6.7. Evidence 7: query scale bounds the method’s applicability

City-wide means can hide local damage, so after freezing all selections we map each query’s expansion difference against Z0 onto a fixed 16×16 grid of equal-sized metric cells over the road network. Hotspots are defined only by

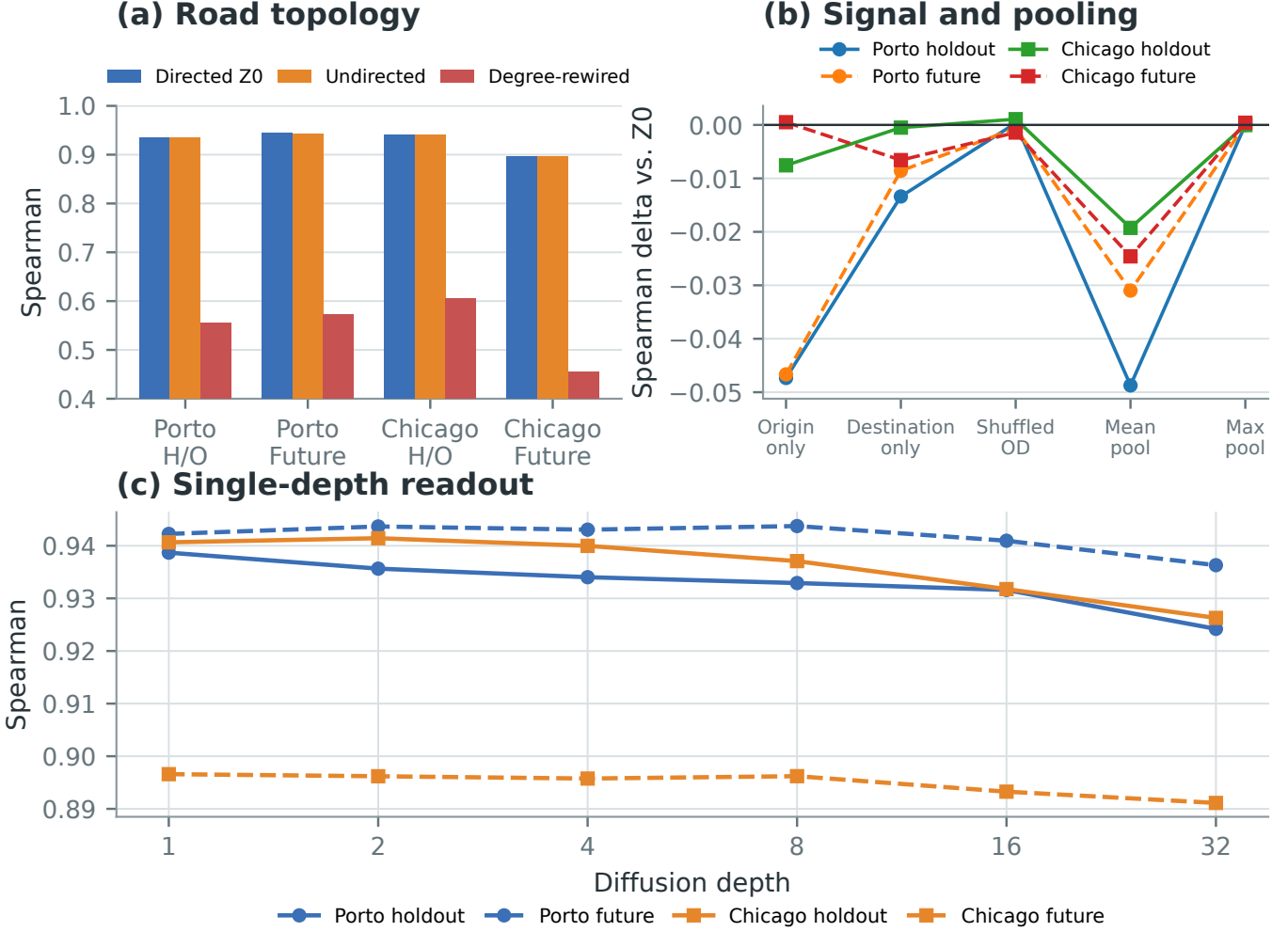


Figure 3: Orthogonal mechanism ablations of Z0. Degree-preserving rewiring damages ranking in both cities and both windows; single-side demand, pooling, and single-depth readouts vary by city and window. All ablations are shown, favorable or not.

endpoint frequency in the history window H ; improved, flat, and degraded cells are all kept on the map. In Porto, BRIDGE-B improves 87.8%/90.0% of active cells in Y/F , with positive mean differences in both hotspot and non-hotspot strata — gains extend outward rather than trading the head for coverage. Chicago’s ratios are 41.2%/43.3%, and neither stratum shows a stable edge over Z0: spatial-coverage improvement does not transfer across cities by default. These maps diagnose mechanisms on the already-unlocked Y/F ; they are not a new out-of-time confirmation.

We then split queries into four equal-frequency groups of 500 by their original-graph expansion count. Table 8 reports BRIDGE-B against the original graph; positive reduction means the overlay expands fewer nodes.

Porto’s heaviest queries each save about 4,800–5,000 expanded nodes. Chicago’s lightest quartile expands only about 240–265 nodes to begin with; the fixed cost of endpoint access and shortcut scanning cannot amortize over so small a search and turns into overhead — yet the heaviest quartile still drops by about 25%. Grouping by straight-line OD span points the same way: Chicago’s longest quartile

falls 24.3%/25.0% in Y/F . The applicability boundary is concrete: selective region materialization suits taxi- and driving-style workloads of mid-to-long distances or high search workload, while short walking- and cycling-style queries may never cover the fixed access cost. What is established here is a query-scale effect, not a transport-mode effect — city graph and mode change together in these two datasets. And since the table compares the whole framework against the original graph, it says nothing about BRIDGE-B beating Z0 on Chicago’s long queries.

7. Discussion

7.1. A division of labor between the deterministic method and its neural upgrades

The experiments support a layered reading. Z0, with no labels and no training, already delivers strong global ranking and online workload gains — the natural default for deployment. BRIDGE wins all 12 comparisons across two cities, three seeds, and two external windows, so the

Table 5: Online expanded-node results under strict $K=18$ disjoint deployment. Negative means fewer expansions than the original graph.

City	Method	Shortcuts	Δ expanded (current)	Δ expanded (future)
Porto	random	7,986	−4.91%	−4.73%
	historical hotspot	7,235	−15.84%	−15.90%
	midpoint proxy	8,307	−17.35%	−17.66%
	Z0	6,712	−17.46%	−17.76%
	BRIDGE (3 seeds)	7,755	−17.59%	−17.80%
Chicago	random	26,011	−4.40%	−4.07%
	historical hotspot	20,093	−17.37%	−17.95%
	midpoint proxy	21,516	−20.30%	−21.08%
	Z0	23,378	−21.47%	−21.92%
	BRIDGE (3 seeds)	20,718	−19.62%	−20.28%

Table 6: $K=18$ performance change under one C++20/O3 implementation. Negative means a reduction.

City	Method	Mean (cur.)	Mean (fut.)	P95 (cur.)	P95 (fut.)	Edge scans (cur.)	Edge scans (fut.)
Porto	Z0	−6.51%	−7.48%	−4.72%	−6.24%	−6.67%	−6.91%
	BRIDGE	−6.41%	−8.90%	−5.56%	−8.96%	−5.85%	−6.05%
	BRIDGE-B	−6.06%	−7.85%	−4.77%	−6.37%	−7.83%	−8.12%
Chicago	Z0	+3.57%	+0.72%	−3.13%	−3.85%	+26.56%	+27.51%
	BRIDGE	+3.03%	+3.02%	−2.24%	−1.33%	+18.58%	+19.26%
	BRIDGE-B	+4.64%	+3.89%	+0.30%	−0.52%	+23.49%	+24.50%

Note: BRIDGE and BRIDGE-B are three-seed means; standard deviations are omitted here and preserved in the released artifacts.

neural residual extracts ordering information the frozen prior genuinely lacks; it earns its place as a global-ranking upgrade with repeated evidence behind it. BRIDGE-B then takes over head and cost control under a fixed budget: its three-seed results show a neural objective can shape a stable deployment set, not just a score vector. Both cities’ online expansions stay within 3% of Z0; Porto buys a small workload improvement while cutting shortcuts, and Chicago trades a small expansion gap for a cheaper index and a stronger ranked head. A production system can pick along this spectrum by its own storage and workload priorities.

7.2. Global order and head selection are different objectives — and why that matters in practice

The second finding worth naming is that global ranking and head selection are not two approximations of one target. Spearman constrains the relative order of all candidates; NDCG@ K and actual deployment care only about the few near the truncation point, where hard-disjoint selection additionally injects overlap and shortcut costs. BRIDGE raising global Spearman 12 times out of 12 while head metrics wander by city and budget is a reproducible dissociation — “ranking better” and “choosing better within budget” come apart under measurement, and a loss aimed

at one does not automatically serve the other. That dissociation is the direct motivation for budget-aware losses and set-level learning.

Nor is the global order a mere average that deployment can ignore. Real systems see K move with storage and preprocessing budgets; high-scoring regions get displaced by spatial overlap, dense shortcuts, or temporary unavailability, and the system must then promote candidates from further down the list. A model tuned to one fixed top- K concentrates resources on today’s hotspots and travels poorly across budget changes; a high-quality full ranking hands every budget the same reusable decision axis. Combined with spatially uniform candidates and historical-OD propagation, that axis weighs busy districts and the rest of the network by their actual demand–gain structure — coverage in the engineering sense, with no claim made about demographic or administrative fairness.

7.3. What BRIDGE-B does and does not establish

The point of BRIDGE-B is not to dress up a few dozen expanded nodes as a speedup. Its value is converting the observed objective dissociation into a training design: soft Spearman maintains the full candidate axis, differentiable top- K protects the truncation point, the budget term prices materialization, and the conflict term anticipates hard-disjoint deployability. Porto’s three seeds agreeing on the

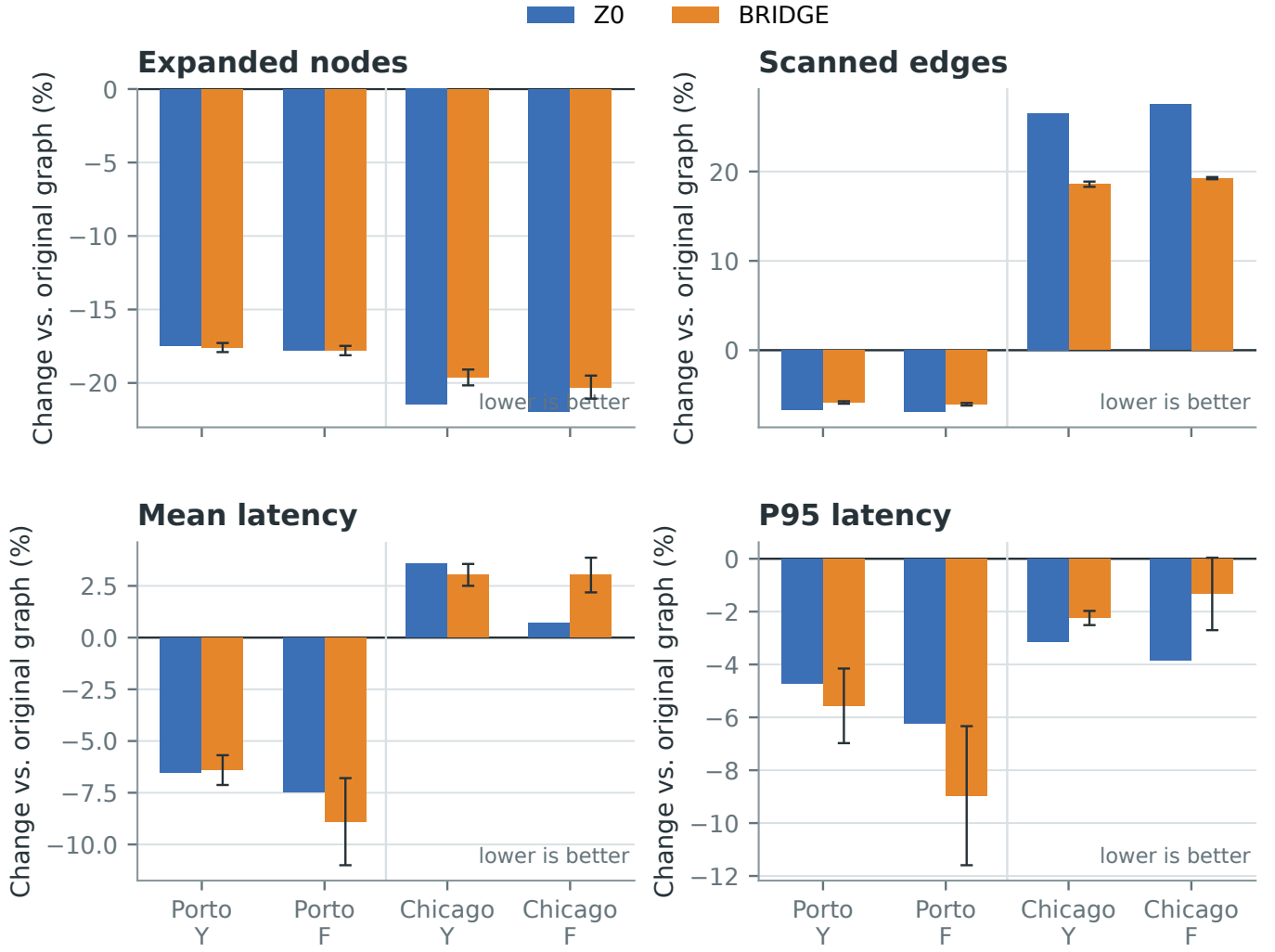


Figure 4: System gains and costs of selective materialization in both cities under one C++20/O3 implementation. Negative means a reduction relative to the original graph. Chicago expands fewer nodes yet scans more edges through dense shortcuts — ranking gains and wall-clock gains are not the same thing.

exact member set — and Chicago’s three agreeing on global, head, and cost direction — says these constraints act on the model rather than luck acting on a seed.

Chicago’s residual expansion gap is itself mechanism information. Shortcut count describes only part of overlay edge density, and single-region labels do not sum to the joint marginal gain of co-materialized regions; endpoint access, region composition, and search order all move the final count. The accurate statement of BRIDGE-B today: a stable budget-aware ranking and cost enhancement that keeps online workload at Z0’s level while lowering shortcuts, with a small real improvement in Porto — and a sub-3% cross-city gap that is neither a replacement claim nor a regression, but the concrete target for a set-level workload surrogate in the next iteration.

7.4. Engineering scope

The query-scale stratification indicates that the framework mainly fits static road-analysis tasks with recurring

queries, tolerance for offline preprocessing, a hard exactness requirement, and enough per-query search work to amortize index access cost. Candidates are fixed at 512 nodes; region scale, label construction cost, and boundary shortcut density all still shape the performance-storage trade.

7.5. Limitations and future work

Porto taxis and Chicago bike share provide cross-city, cross-mode evidence, but not strong demand shifts, dynamic edge weights, or a wider range of network densities. Temporal isolation, spatial overlap-group isolation, frozen multi-seed evaluation, and per-query distance checks lower the risk of bias, but the budget-aware extension was developed after the main experiments had already unlocked the current and future windows; its cross-seed stability is confirmed, while genuinely new out-of-time conclusions should come from later windows or unseen cities. Query mode and city graph structure change together in our data,

Table 7: BRIDGE-B three-seed frozen ranking and $K=18$ online results. $\Delta Y/\Delta F$ are per-query expanded-node differences against Z0; negative is better.

City	Seed	Holdout Spearman	NDCG@5	Shortcuts	ΔY	ΔF
Porto	42	0.937893	0.930720	6,644	-75.478	-76.201
	43	0.939052	0.930720	6,644	-75.478	-76.201
	44	0.938122	0.930720	6,644	-75.478	-76.201
Chicago	42	0.942985	0.994012	23,104	+54.250	+62.103
	43	0.942172	0.994012	23,104	+54.250	+62.103
	44	0.942121	0.959441	22,581	+89.340	+91.889

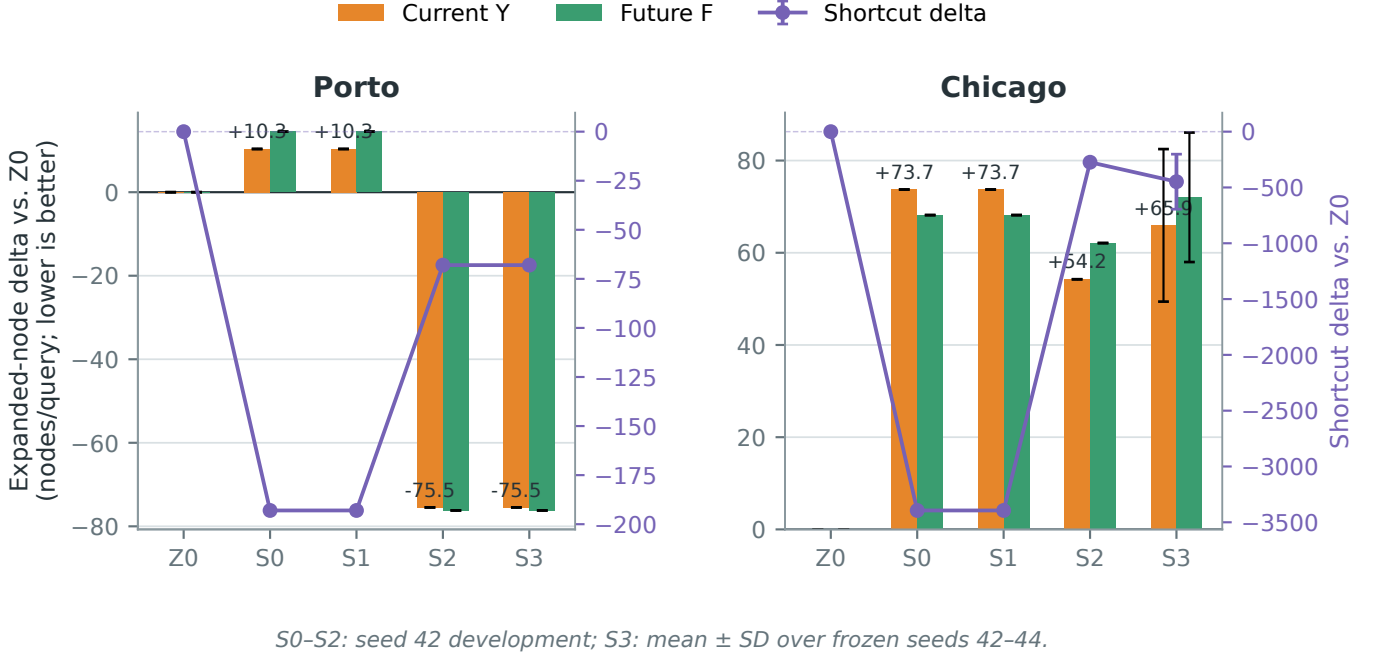


Figure 5: BRIDGE-B’s real deployment trajectory from S0 to S3. S0-S2 are seed-42 development stages; S3 shows the frozen protocol over seeds 42/43/44 with means and population standard deviations. Porto improves with zero variance; Chicago stably cuts shortcuts and holds the head; both cities stay within 3% of Z0 in online expansions.

so we claim a query-scale effect only, and do not read the two-city difference as a causal transport-mode effect.

Natural next steps: set-level expansion and edge-scan surrogates, boundary shortcut sparsification, dynamic demand updates, and more cities. Because BRIDGE-B’s development stage has seen the current Y/F , its next freeze must bring a new future window or an unseen city rather than more weight tuning on the same data. For dense-boundary graphs like Chicago, moving edge scans and endpoint access directly into the selection objective looks more promising than pushing the same top- K gain weight further.

8. Conclusion

Around the main question — can historical demand guide exact region materialization under a limited budget — this paper gives an affirmative but bounded answer. Parameter-free propagation forms a strong region-value ranking on its own; the neural residual then improves the

full ranking consistently in 12 of 12 cross-city, cross-seed, cross-window paired comparisons; and budget-aware training folds head gain, index cost, and region conflicts into one decision, obtaining a stable online improvement in Porto while holding Chicago’s workload gap to the strong prior within 3%. All online queries preserve exact distances, which means artificial intelligence can serve as an offline resource allocator alongside an existing shortest-path service instead of taking over the queries themselves. The experiments equally show that the full ranking, the budget head, and real deployment gain are different objectives: the first supports budget changes, conflict-driven elimination, and network-wide stand-by candidates, while the latter two still demand set-level online verification. The framework best fits mid-to-long-distance or high-workload queries that can amortize access costs, and it can join existing engineering systems through asynchronous shadow sampling, offline statistics, and periodic materialization.

Table 8: BRIDGE-B across query-workload scales. Q1/Q4 are the lowest/highest quartiles of original-graph expansions.

City	Window	Q1 baseline exp.	Q1 reduction	Q4 baseline exp.	Q4 reduction	Q4 nodes saved
Porto	Y	661	+3.98%	32,015	+15.74%	5,039
	F	727	+9.18%	30,464	+15.90%	4,843
Chicago	Y	265	−97.11%	9,663	+24.92%	2,408
	F	238	−111.08%	9,212	+25.72%	2,370

Appendix A. Deep-training stability diagnosis

The early 32-layer network sat still for about 22 epochs, then changed abruptly. Step-by-step diagnosis found that FP16’s default dynamic loss scaling caused the early optimizer steps to be skipped outright; under FP32/BF16, individual gradient elements stayed finite, but the FP32 sum of squares inside the global-norm computation overflowed, and the clipping coefficient collapsed toward zero. Backward hooks showed amplification from about 10^{-6} at layer 32 to about 10^{19} at layer 1. The strict identity path in Equation (8) is what stabilized deep training. The old pairwise loss was later found to be misaligned with Spearman, and the soft Spearman protocol was frozen in its place. The diagnosis separates numerical trainability from task-objective validity; it does not affect the main text’s evidence structure around the materialization question.

Appendix B. Frozen configuration summary

Appendix C. Spatial benefit heterogeneity maps

Appendix D. Glossary

Region materialization Pre-encoding the exact distances inside a candidate region as boundary connections, which replace the region’s interior nodes in the query graph.

Region gain The mean reduction in expanded nodes for exact queries, relative to the original graph, when a single candidate region is materialized.

Full ranking A continuous priority over all candidate regions, rather than only the selected set at one fixed budget.

Budget head The candidates near the ranking truncation point, and their cumulative gain, under a given region-count or index-cost limit.

Strictly disjoint deployment No two co-materialized regions share a road node.

Shadow label sampling Asynchronously mirroring a small share of real queries to offline workers that never affect production responses, in order to measure region gain.

Data availability

Raw data come from the public sources cited in the text. The public scope and permanent links for the processed

data, fixed splits, and labels will be added after the license audit is completed before submission.

Code availability

An anonymous reproduction repository and its archival link will be added before submission.

Declaration of generative AI and AI-assisted technologies

During this research and the preparation of the manuscript, the authors used OpenAI Codex to assist with writing, refactoring, and debugging parts of the research code, tests, and experiment scripts, and with the structure and language of the article. All code and text were reviewed, tested, verified, and edited by the authors as needed; the experimental data come from real program executions, and method selection, result interpretation, and scientific conclusions were made by the authors. The authors take full responsibility for the content of the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

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Jiawen Wang: _____

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Table B.9: Key frozen configurations of the formal experiments.

Item	Frozen value
Time windows	$H = [0, 0.35)$, $Y = [0.35, 0.70)$, $F = [0.70, 1.00)$
Candidates	1,200 per city, 512 nodes each, candidate seed 42
Spatial isolation	Jaccard-0.50 overlap components, 840/180/180
Label queries	2,000 per city, window, and candidate; sampling seed 42; endpoint cache off
Z0	forward/reverse lazy mean, depths 1/2/4/8/16/32, region mean+max
BRIDGE	32-layer two-tower, hidden dim 32, residual scale 0.01, 418,183 parameters
Optimization	CUDA FP32, lr 5×10^{-3} , weight decay 0, 40 epochs, no scheduler
Model selection	seeds 42/43/44, validation Spearman, $\alpha \in \{0, 0.125, 0.25, 0.5, 1\}$
BRIDGE-B	same towers and frozen Z0, 12 epochs, top- K /budget/conflict weights 1.0/1.0/0.10
BRIDGE-B selection	max validation top- K gain, shortcuts $\leq Z0$, Spearman tolerance 0.005
Deployment	hard-disjoint, $K = 5/10/18$; main system tables report $K = 18$
C++ timing	C++20/O3, one pinned core, 2 warm-up + 10 timed rounds, 2,000 queries per cell

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Porto: spatial deployment benefit relative to Z0

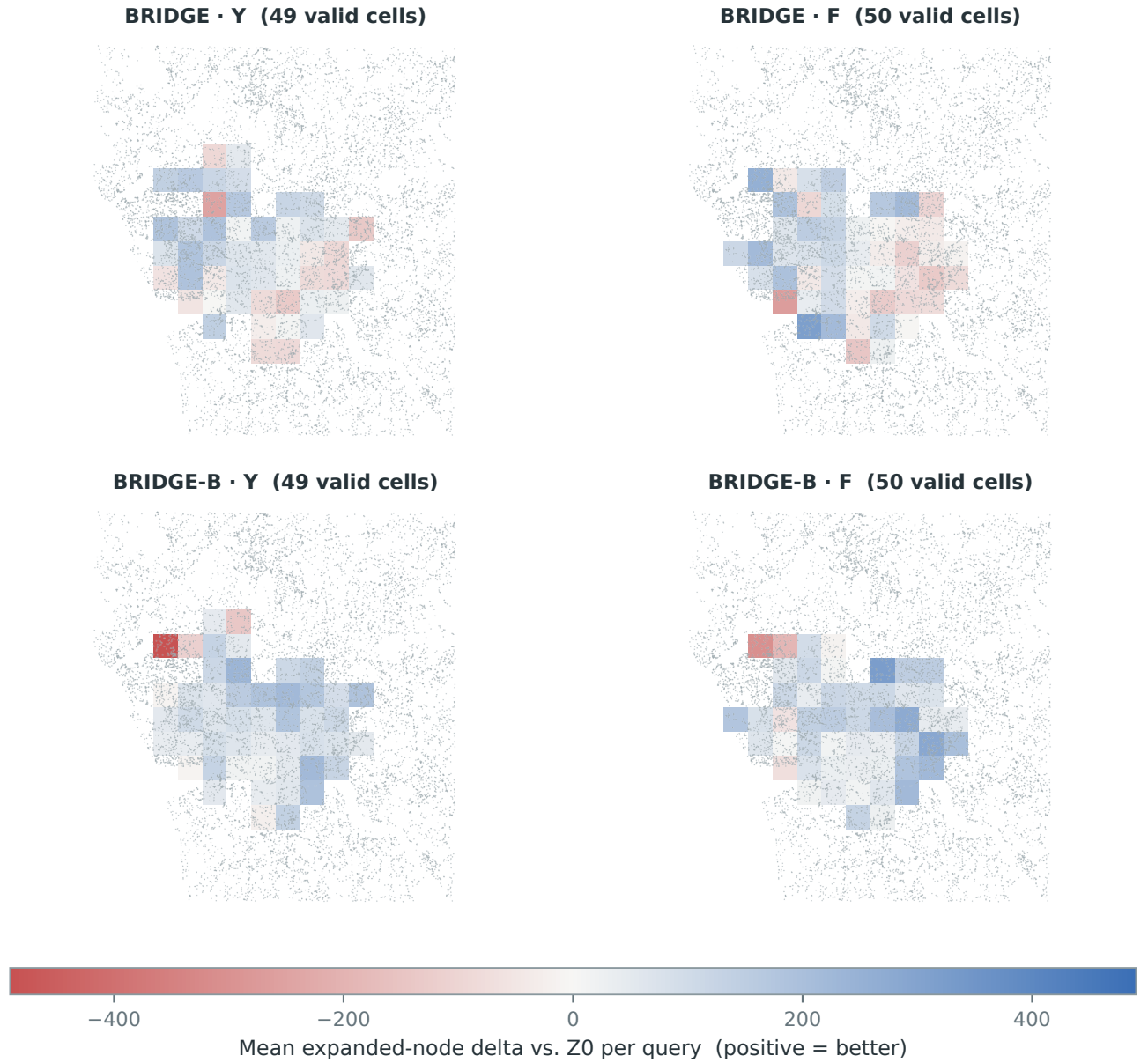


Figure C.6: Porto: per-query expanded-node differences of BRIDGE and BRIDGE-B against Z0 in Y/F , on the spatial grid. Blue means the neural method expands fewer nodes; BRIDGE-B's positive coverage is broad in both windows.

Chicago: spatial deployment benefit relative to Z0

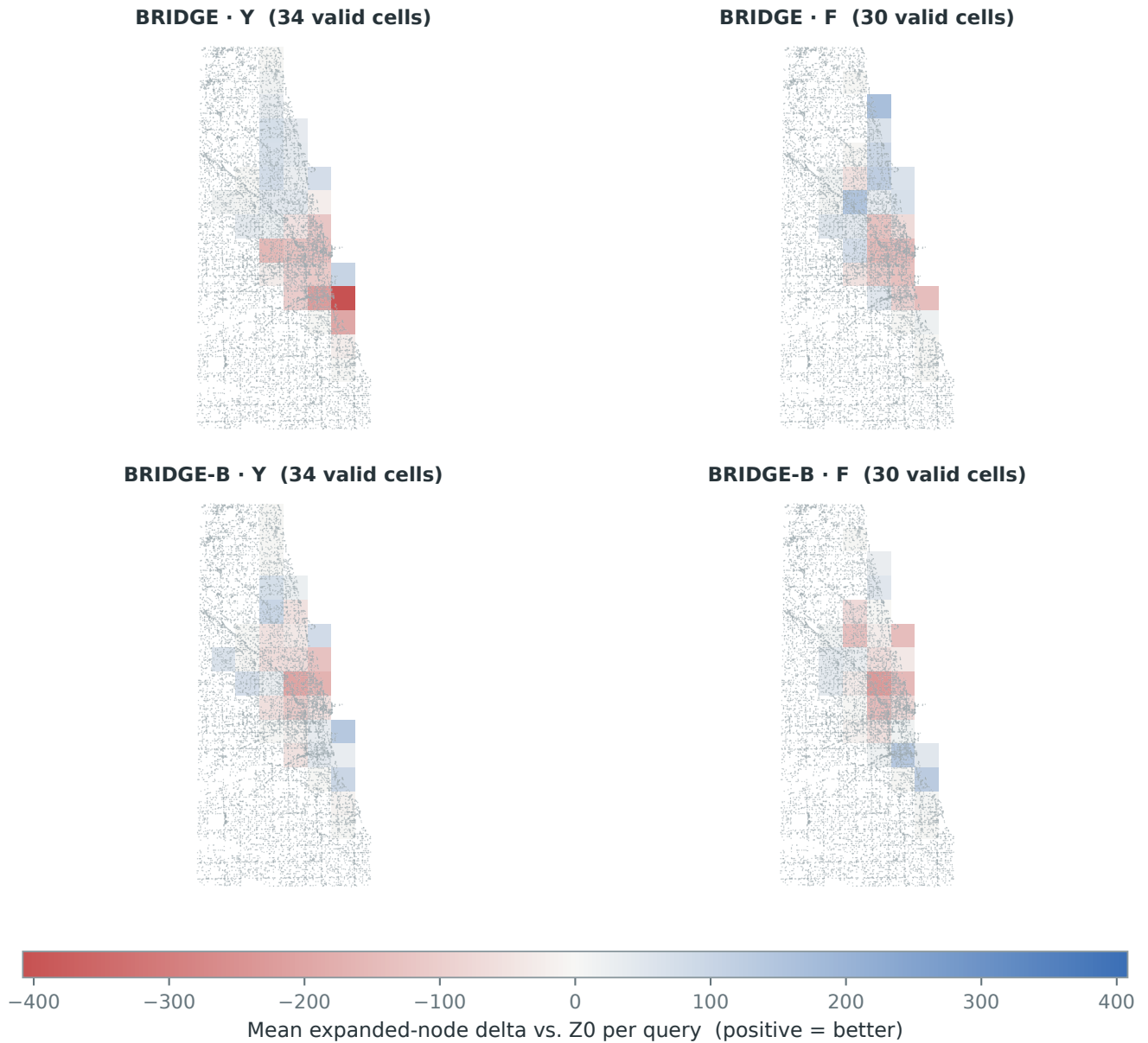


Figure C.7: Chicago: per-query expanded-node differences of BRIDGE and BRIDGE-B against Z0 in Y/F . Positive and negative cells coexist — local improvements do not add up to a city-wide advantage.